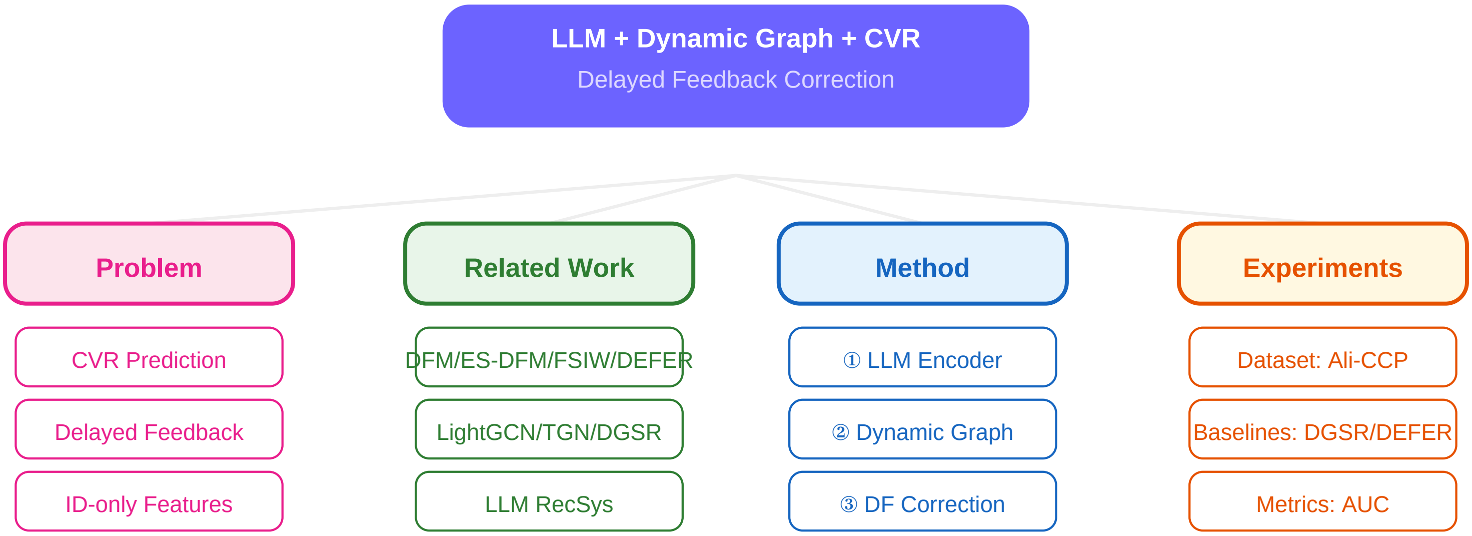
Dissertation Proposal

LLM-Enhanced Dynamic Graph Networks for Conversion Rate Prediction

Liu Yize | UCL MSc KIDS | June 2026

RESEARCH OVERVIEW



RESEARCH QUESTIONS

RQ1

LLM embeddings vs. ID embeddings in CVR prediction performance

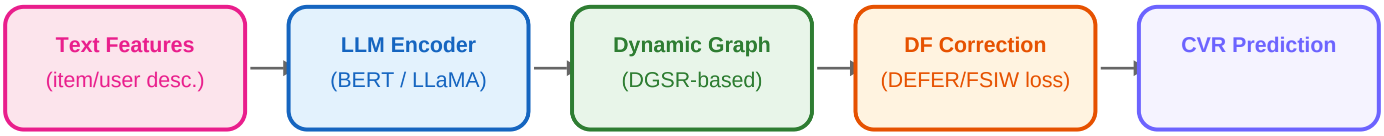
RQ2

Do LLM semantics reduce delayed feedback false negatives?

RQ3

Computational cost vs. accuracy gain trade-off?

PROPOSED ARCHITECTURE



Key Talking Points

Supervisor Meeting — June 2026

1. Literature Review — What I've Read

- Delayed Feedback (4 papers): DFM, ES-DFM, FSIW, DEFER
  → Core insight: training labels are unreliable due to conversion delay; approaches range from EM-based delay modelling (DFM) to distributional correction (FSIW/DEFER)
- Graph Neural Networks (3 papers): LightGCN, TGN, DGSR
  → DGSR is my base architecture: dynamic graph with cross-user collaborative signals, long-term + short-term preference modelling
- LLM for RecSys (in progress): BERT4Rec, UniSRec, RLMRec, TALLRec — reading this week

2. Research Gap & Motivation

- Existing CVR models (including DGSR) rely on ID-based embeddings — no semantic understanding
- Delayed feedback correction methods do not leverage semantic item/user representations
- No prior work integrates LLM embeddings into dynamic graph networks for CVR with delayed feedback
  → My dissertation bridges all three: LLM + Dynamic Graph + Delayed Feedback Correction

3. Proposed Methodology

- Layer 1 — LLM Encoder: encode item/user text descriptions → semantic embeddings
  (Replace ID embeddings in DGSR with LLM-derived representations)
- Layer 2 — Dynamic Graph: DGSR-style neighbour aggregation on the semantic graph
  (Long-term preference via order-aware attention + short-term via last-item attention)
- Layer 3 — Delayed Feedback Correction: modified training objective
  (Drawing from DEFER/FSIW to correct false-negative labels during training)

4. Planned Experiments

- Dataset: Evaluating Ali-CCP (Alibaba, click+conversion) vs. Criteo Conversion — to confirm next week
- Baselines: DGSR (graph w/o LLM), DEFER (delayed feedback w/o graph), LightGCN, DFM
- Metrics: AUC, NLL (model fit), Inference Latency (computational trade-off)
- Ablation: LLM only | Graph only | DF-correction only | Full model

5. Timeline

This week	Complete LLM RecSys paper review (4 papers)
Next week	Finalise dataset selection & begin preprocessing
Week 3–4	Draft Methodology chapter + implement baselines
Month 2	Run initial experiments, collect results
Month 3	Full analysis, write-up, dissertation submission